

Corporate Bankruptcy Prediction

A Comparative Study of Decision Tree and Gradient Boosting Classifiers

Mini Project Report

Domain: Finance / Business Analytics

Techniques Used: Decision Tree, Gradient Boosting (Ensemble Learning)

Abstract

Corporate bankruptcy prediction is a critical task for banks, investors, and regulators, as early identification of financially distressed firms enables timely risk mitigation. This project builds and compares two supervised machine learning models — a Decision Tree classifier and a Gradient Boosting ensemble classifier — to predict whether a company is likely to go bankrupt based on its financial ratios. The models were trained and evaluated on a publicly available dataset of Taiwanese listed companies, comprising financial statement data for 6,819 firms across 95 financial indicators. Model performance was assessed using Accuracy, Precision, Recall, and F1-Score, with particular attention paid to the class imbalance inherent in bankruptcy data. The Decision Tree model additionally provided interpretable insight into which financial ratios most strongly influence bankruptcy risk. The findings and limitations of this study are discussed, along with a proposed direction for extending the framework to Indian corporate financial data.

1. Introduction

Predicting whether a company will become insolvent is a long-standing problem in financial analytics. Traditional approaches relied on statistical techniques such as Altman's Z-Score and discriminant analysis. With the growth of machine learning, tree-based classifiers such as Decision Trees and ensemble boosting methods have become popular for this task because they can model complex, non-linear relationships among financial ratios while remaining reasonably interpretable.

This project applies and compares two such models — a single Decision Tree and a Gradient Boosting ensemble — to classify companies as bankrupt or non-bankrupt using their reported financial ratios. The objective is not only to measure predictive performance, but also to understand which financial indicators are most informative for early detection of financial distress.

1.1 Objectives

- To preprocess and analyze a real-world corporate financial dataset.
- To build a baseline Decision Tree classifier for bankruptcy prediction.
- To build a Gradient Boosting classifier and compare its performance against the baseline.
- To identify the financial ratios most predictive of bankruptcy.
- To critically evaluate model performance under class imbalance using appropriate metrics.

1.2 Note on Dataset Scope

This study uses the publicly available Taiwanese Economic Journal bankruptcy dataset, as a comparable open, ready-to-use dataset of Indian corporate bankruptcies with financial ratio data at the firm level is not currently available in the public domain. Existing Indian studies typically rely on small, private, or restricted-access samples (fewer than 200 firms). Extending this framework to Indian company data is proposed as future work in Section 6.

2. Methodology

2.1 Dataset

The dataset used is the **Taiwanese Bankruptcy Prediction Dataset**, sourced from the UCI Machine Learning Repository (originally collected from the Taiwan Economic Journal, 1999–2009). It contains **6,819 company records** described by **95 financial ratio features** (e.g. profitability, liquidity, leverage, turnover ratios), with a binary target variable '**Bankrupt?**' indicating whether the company went bankrupt.

Class	Number of Companies	Percentage
Non-Bankrupt (0)	6,599	96.8%
Bankrupt (1)	220	3.2%
Total	6,819	100%

Table 1: Class distribution showing significant class imbalance (only 3.2% bankrupt cases).

This strong class imbalance is a defining challenge of the bankruptcy prediction problem: a naive model that always predicts 'non-bankrupt' would already achieve 96.8% accuracy without learning anything useful. This motivates the use of Precision, Recall, and F1-Score — not accuracy alone — as the primary evaluation metrics in this study.

2.2 Data Preprocessing

- Loaded the dataset using the pandas library and verified there were no structural inconsistencies.
- Separated the target variable ('Bankrupt?') from the 95 predictor features.
- Split the data into training (80%) and testing (20%) sets using stratified sampling, to preserve the same bankrupt/non-bankrupt ratio in both sets.

2.3 Models Used

Decision Tree Classifier

A single Decision Tree (max depth = 5) was trained as an interpretable baseline model. Decision Trees split the data recursively based on feature thresholds that best separate the two classes, producing a set of human-readable rules.

Gradient Boosting Classifier

A Gradient Boosting ensemble (100 estimators) was trained as the comparison model. Boosting builds a sequence of shallow decision trees, where each new tree focuses on correcting the errors of the previous ones, typically improving predictive accuracy over a single tree at the cost of some interpretability.

2.4 Evaluation Metrics

- **Accuracy** — overall proportion of correct predictions.
- **Precision** — of all companies predicted bankrupt, how many actually were.
- **Recall** — of all companies that actually went bankrupt, how many were correctly identified.
- **F1-Score** — harmonic mean of Precision and Recall, useful under class imbalance.

3. Results

3.1 Model Performance Comparison

Metric	Decision Tree	Gradient Boosting
Accuracy	96.85%	96.92%
Precision	51.43%	53.85%
Recall	40.91%	31.82%
F1-Score	45.57%	40.00%

Table 2: Performance comparison of Decision Tree vs Gradient Boosting on the test set.

Both models achieved similarly high accuracy (~97%), which, as discussed in Section 2.1, is expected given the class imbalance and is not by itself a meaningful indicator of model quality. Looking at Recall and F1-Score — the more informative metrics here — the **Decision Tree outperformed Gradient Boosting**, correctly identifying 40.9% of actual bankrupt companies compared to 31.8% for Gradient Boosting.

This result is notable because Gradient Boosting is generally expected to outperform a single Decision Tree. A likely explanation is that boosting algorithms, without explicit class-weighting or resampling, tend to optimize for overall accuracy and can become biased toward the majority (non-bankrupt) class when the minority class is severely under-represented, as is the case here. This is a well-documented limitation of boosting methods on imbalanced datasets and represents a meaningful, genuine finding of this study rather than an implementation error.

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Figure 1: Confusion matrices for Decision Tree (left) and Gradient Boosting (right).

3.2 Feature Importance

The Decision Tree model was also used to identify which financial ratios contributed most to its predictions. The top 10 most important features are listed below:

Rank	Financial Ratio	Importance
1	Net Value Growth Rate	0.364
2	Interest-bearing Debt Interest Rate	0.153
3	Borrowing Dependency	0.104
4	Continuous Interest Rate (after tax)	0.087
5	ROA(B) before interest & depreciation, after tax	0.053
6	Net Worth Turnover Rate (times)	0.043

7	Cash Flow to Sales	0.036
8	Operating Gross Margin	0.033
9	No-credit Interval	0.026
10	Fixed Assets Turnover Frequency	0.022

Table 3: Top 10 most important financial ratios identified by the Decision Tree model.

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Figure 2: Bar chart of the top 10 important features.

The dominance of **Net Value Growth Rate** and debt-related ratios (interest-bearing debt interest rate, borrowing dependency) aligns with financial theory: companies with shrinking net value and heavy reliance on borrowed capital are structurally more vulnerable to insolvency.

4. Discussion

The results highlight an important practical lesson in applied machine learning: a more sophisticated algorithm (Gradient Boosting) does not automatically outperform a simpler one (Decision Tree) on every dataset, especially under severe class imbalance. In a real-world deployment for bank credit risk assessment, Recall is arguably the more important metric, since failing to identify a company that will actually go bankrupt (a false negative) is typically far costlier than incorrectly flagging a healthy company for review (a false positive). By this criterion, the Decision Tree's higher Recall makes it the more practically useful model in this experiment, despite Gradient Boosting's marginally higher raw accuracy.

A key limitation of this study is that no class-balancing technique (such as SMOTE or class-weighting) was applied. Applying such techniques would likely improve Recall for both models, particularly Gradient Boosting, and is a natural next step.

5. Conclusion

This project successfully built and compared a Decision Tree and a Gradient Boosting classifier for predicting corporate bankruptcy using financial ratio data. While both models achieved high overall accuracy, the Decision Tree demonstrated better Recall and F1-Score, making it more effective at identifying at-risk companies in this imbalanced dataset. The Decision Tree also provided interpretable insight into the financial ratios most strongly associated with bankruptcy risk, with net value growth and debt-related ratios emerging as the strongest predictors. These findings underline the importance of selecting evaluation metrics and modelling techniques appropriate to the structure of the data, rather than assuming that ensemble methods will always outperform simpler models.

6. Future Scope

- Apply class-imbalance handling techniques (SMOTE, class-weighting, undersampling) and re-evaluate both models.
- Extend this framework to Indian corporate financial data (e.g. NCLT/IBC insolvency records or listed-company financial statements) once a suitable public dataset becomes available, enabling region-specific bankruptcy prediction.
- Experiment with additional ensemble methods such as XGBoost, LightGBM, and Random Forest for further comparison.
- Incorporate macroeconomic indicators (interest rates, GDP growth) alongside firm-level financial ratios.

7. Tools and Technologies Used

- **Language:** Python 3.10
- **Libraries:** pandas, numpy, scikit-learn, matplotlib, seaborn
- **Version Control:** Git and GitHub
- **Dataset Source:** UCI Machine Learning Repository / Kaggle (Taiwanese Bankruptcy Prediction Dataset)